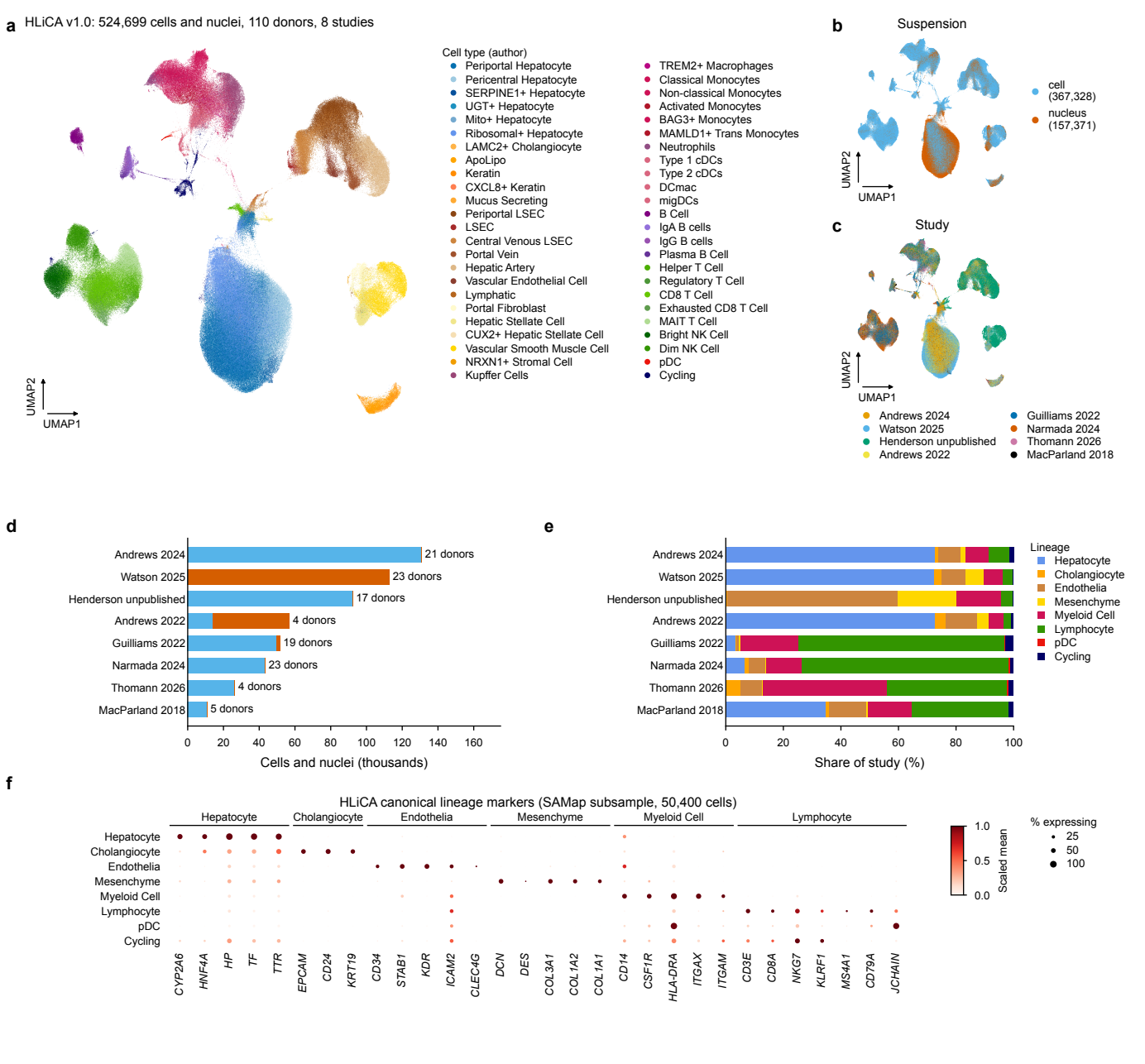

a
HLiCA v1.0: 524,699 cells and nuclei, 110 donors, 8 studies
UMAP2
UMAP1
Cell type (author)
Periportal Hepatocyte
Pericentral Hepatocyte
SERPINE1+ Hepatocyte
UGT+ Hepatocyte
Mito+ Hepatocyte
Ribosomal+ Hepatocyte
LAMC2+ Cholangiocyte
ApoLipo
Keratin
CXCL8+ Keratin
Mucus Secreting
Periportal LSEC
LSEC
Central Venous LSEC
Portal Vein
Hepatic Artery
Vascular Endothelial Cell
Lymphatic
Portal Fibroblast
Hepatic Stellate Cell
CUX2+ Hepatic Stellate Cell
Vascular Smooth Muscle Cell
NRXN1+ Stromal Cell
Kupffer Cells
TREM2+ Macrophages
Classical Monocytes
Non-classical Monocytes
Activated Monocytes
BAG3+ Monocytes
MAMLD1+ Trans Monocytes
Neutrophils
Type 1 cDCs
Type 2 cDCs
DCmac
migDCs
B Cell
IgA B cells
IgG B cells
Plasma B Cell
Helper T Cell
Regulatory T Cell
CD8 T Cell
Exhausted CD8 T Cell
MAIT T Cell
Bright NK Cell
Dim NK Cell
pDC
Cycling
b
Suspension
cell (367,328)
nucleus (157,371)
UMAP2
UMAP1
c
Study
UMAP2
UMAP1
Andrews 2024
Watson 2025
Henderson unpublished
Andrews 2022
Guilliams 2022
Narmada 2024
Thomann 2026
MacParland 2018
d
Andrews 2024
Watson 2025
Henderson unpublished
Andrews 2022
Guilliams 2022
Narmada 2024
Thomann 2026
MacParland 2018
21 donors
23 donors
17 donors
4 donors
19 donors
23 donors
4 donors
5 donors
0
20
40
60
80
100
120
140
160
Cells and nuclei (thousands)
e
Andrews 2024
Watson 2025
Henderson unpublished
Andrews 2022
Guilliams 2022
Narmada 2024
Thomann 2026
MacParland 2018
0
20
40
60
80
100
Share of study (%)
Lineage
Hepatocyte
Cholangiocyte
Endothelia
Mesenchyme
Myeloid Cell
Lymphocyte
pDC
Cycling
f
HLiCA canonical lineage markers (SAMap subsample, 50,400 cells)
Hepatocyte
Cholangiocyte
Endothelia
Mesenchyme
Myeloid Cell
Lymphocyte
CYP2A6
HNF4A
HP
TF
TTR
EPCAM
CD24
KRT19
CD34
STAB1
KDR
ICAM2
CLEC4G
DCN
DES
COL3A1
COL1A2
COL1A1
CD14
CSF1R
HLA-DRA
ITGAX
ITGAM
CD3E
CD8A
NKG7
KLRF1
MS4A1
CD79A
JCHAIN
Scaled mean
1.0
0.5
0.0
% expressing
25
50
100